

--TASK 1 CREATE NEW DATABASE

The test_db was created to the default tablespace pg_default.

```
--TASK 1
CREATE DATABASE test_db;

select d.oid, d.datname, d.datistemplate, d.dataallowconn, t.spcname
from pg_database d
join pg_tablespace t on t.oid = d.dattablespace
```

	oid	datname	datistemplate	dataallowconn	spcname
1	5	postgres	[]	[v]	pg_default
2	17,337	test_db	[]	[v]	pg_default
3	1	template1	[v]	[v]	pg_default
4	4	template0	[v]	[]	pg_default

--TASK 2 CREATE NEW TABLESPACE

The new tablespace was created and test_db was moved to it.

```
*-<postgres> Script-7 X

CREATE DATABASE test_db;

select d.oid, d.datname, d.datistemplate, d.dataallowconn, t.spcname
from pg_database d
join pg_tablespace t on t.oid = d.dattablespace

--TASK 2 CREATE NEW TABLESPACE

CREATE TABLESPACE mytablespace LOCATION 'C:\Program Files\PostgreSQL\18\data\tblspc_test\

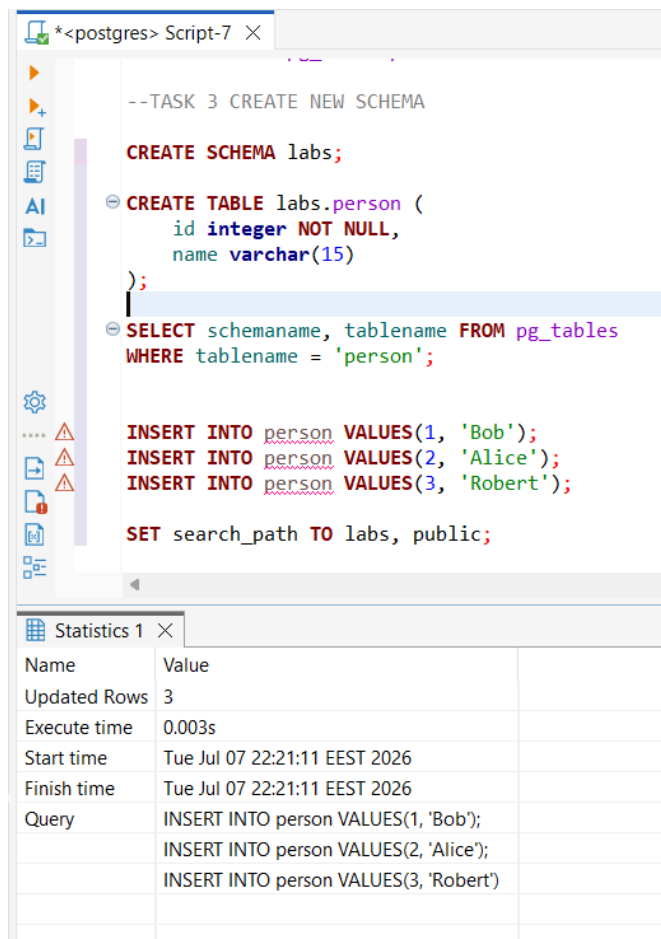
ALTER DATABASE test_db SET TABLESPACE mytablespace

select * from pg_tablespace
```

	oid	datname	datistemplate	dataallowconn	spcname
1	5	postgres	[]	[v]	pg_default
2	1	template1	[v]	[v]	pg_default
3	4	template0	[v]	[]	pg_default
4	17,337	test_db	[]	[v]	mytablespace

--TASK 3 CREATE NEW SCHEMA

The schema 'lab' was created and the search route was directed to it.
Initial data inserted.



The screenshot shows a PostgreSQL script editor window titled '*<postgres> Script-7'. The script contains the following SQL commands:

```
--TASK 3 CREATE NEW SCHEMA

CREATE SCHEMA labs;

CREATE TABLE labs.person (
  id integer NOT NULL,
  name varchar(15)
);

SELECT schemaname, tablename FROM pg_tables
WHERE tablename = 'person';

INSERT INTO person VALUES(1, 'Bob');
INSERT INTO person VALUES(2, 'Alice');
INSERT INTO person VALUES(3, 'Robert');

SET search_path TO labs, public;
```

Below the script editor, a 'Statistics 1' window is open, displaying the following data:

Name	Value
Updated Rows	3
Execute time	0.003s
Start time	Tue Jul 07 22:21:11 EEST 2026
Finish time	Tue Jul 07 22:21:11 EEST 2026
Query	INSERT INTO person VALUES(1, 'Bob'); INSERT INTO person VALUES(2, 'Alice'); INSERT INTO person VALUES(3, 'Robert')

-- TASK 4 INVESTIGATE MVCC

Separate transaction blocks to insert, update, and delete rows were performed
The transaction flow and records positions were reflected on t_xmin and t_xmax values.

person 1	person 2	Results 3	Results 5	person 5	Statistics 5	Results 6	Results 8 X
SELECT t_xmin, t_xmax, t_ctid, tuple_data_split('la' Enter a SQL expression to filter results (use Ctrl+Space)							
	t_xmin	t_xmax	t_ctid	tuple_data_split			
▶ 1	954	0	(0,1)	▶ \x01000000 [+1]			
▶ 2	954	956	(0,5)	▶ \x02000000 [+1]			
▶ 3	954	957	(0,3)	▶ \x03000000 [+1]			
▶ 4	955	0	(0,4)	▶ \x04000000 [+1]			
▶ 5	956	0	(0,5)	▶ \x02000000 [+1]			
▶ 6	958	959	(0,6)	▶ \xe7030000 [+1]			

--TASK 5 INVESTIGATE VACUUM

VACUUM labs.person command identified the 3 dead row versions and marked them reusable.

The 'Sarah' record landed on the 'new' row.

VACUUM FULL labs.person command built a completely empty data file and copied *only* the live rows ("Bob", "John", updated "Alex" and "Sarah") over to it. The old file was deleted.

person 1	person 2	Results 3	Results 5	person 5	Results 6
SELECT t_xmin, t_xmax, t_ctid, tuple_data_split('la' Enter a SQL expression to filter results (use Ctrl					
	t_xmin	t_xmax	t_ctid	tuple_data_split	
▶ 1	954	0	(0,1)	▶ \x01000000 [+1]	
▶ 2	960	0	(0,2)	▶ \x05000000 [+1]	
▶ 3	955	0	(0,3)	▶ \x04000000 [+1]	
▶ 4	956	0	(0,4)	▶ \x02000000 [+1]	